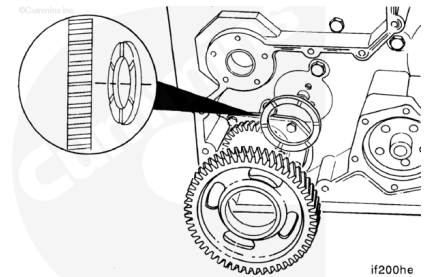


Trainings ▾

Install

⚠ CAUTION ⚠

The grooved side of the rear thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.

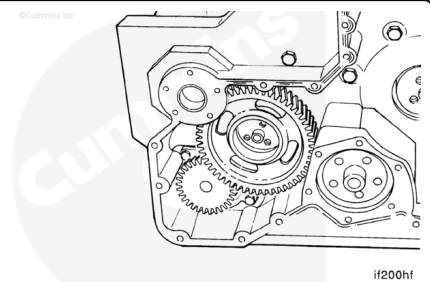


Use Lubriplate™ 105, or equivalent, to lubricate the thrust bearings and idler gear shaft.

Install the rear thrust bearing.

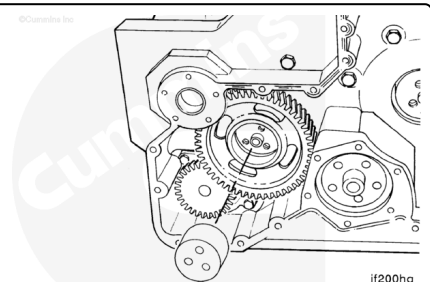
Install the idler gear without the shaft.

Align the inside diameter of the rear thrust bearing with the inside diameter of the idler gear.



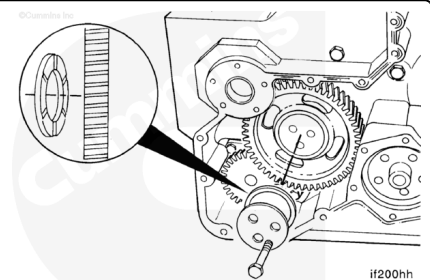
The shaft **must** pass through the gear and pilot into the rear thrust bearing. After installation, the shaft **must** protrude **only** slightly more than the thickness of the front thrust bearing. Excessive shaft protrusion beyond the thickness of the front thrust bearing indicates the shaft is **not** properly piloted into the rear thrust bearing.

Install the shaft into the gear bore.



⚠ CAUTION ⚠

The grooved side of the front thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.



Install the front thrust bearing.

Install the gear retainer.

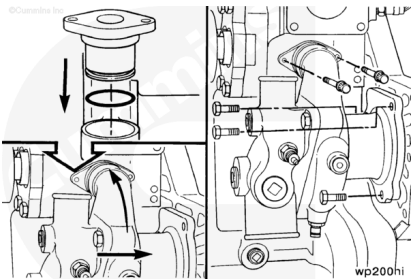
Install the capscrews and tighten.

Torque Value:

1. 61 n•m [45 ft-lb]

2. Rotate 60 degrees

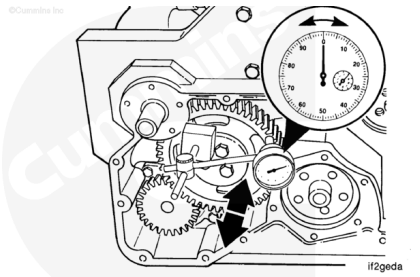
Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)



The following idler gear end clearance and backlash checks apply to all three idler gears. The same MINIMUM and MAXIMUM values apply, also.

Use a dial indicator gauge with a magnetic base to measure the idler gear end clearance.

Place the contact tip of the gauge against the face of the idler gear.



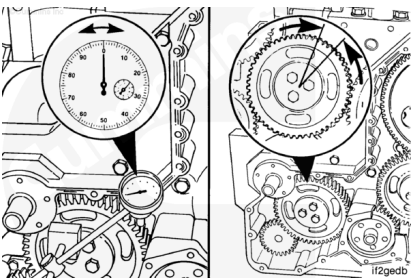
Idler Gear End Clearance

mm		in
0.30	MIN	0.012
0.53	MAX	0.021

Use a dial indicator gauge with a magnetic base to measure the idler gear backlash.

Place the contact tip of the gauge against a tooth on the idler gear.

Do **not** allow the mating gears to move while measuring the backlash.



Idler Gear Backlash

mm		in
0.08	MIN	0.003
0.38	MAX	0.015

Last Modified: 21-Jan-2009
